Chapter 9: Model-Based Clustering using Gaussian Mixture Models

1. Introduction

Gaussian Mixture Models (GMM) provide a probabilistic approach to clustering that assigns soft (probabilistic) labels to observations. Unlike K-means (which uses hard assignments), GMM quantifies uncertainty in cluster membership through cluster assignment probabilities. BIC and AIC information criteria help determine the optimal number of clusters.

2. Dataset & Variables

Network centrality measures from a Moodle MOOC course (n=445 nodes). Features: InDegree, OutDegree, Closeness_total, Betweenness, Eigen, Diffusion.degree, Coreness, Cross_clique_connectivity. Centrality measures capture each student's position and role in the course interaction network.

3. Methods

Features were standardized (zero mean, unit variance). Gaussian Mixture Models were fit for k=1 to k=10 components. BIC (Bayesian Information Criterion) and AIC (Akaike Information Criterion) were computed for each k. The optimal model (k=9) was selected based on minimum BIC. Soft cluster probabilities were extracted, and results compared with K-means clustering.

4. Results

Model Selection (BIC/AIC):

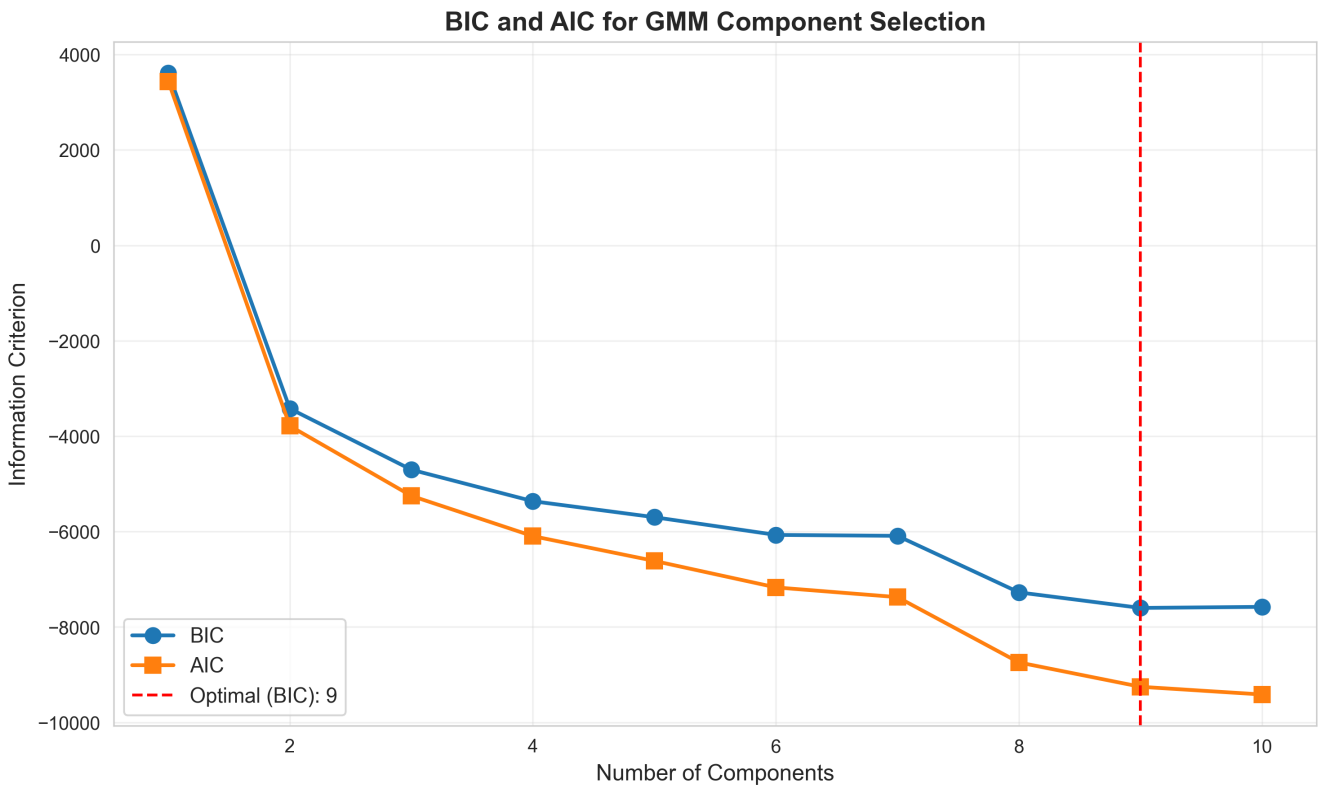

Fig 1: BIC and AIC scores across different numbers of components. BIC achieves minimum at k=9, indicating the optimal cluster count for this dataset.

Assignment Probabilities:

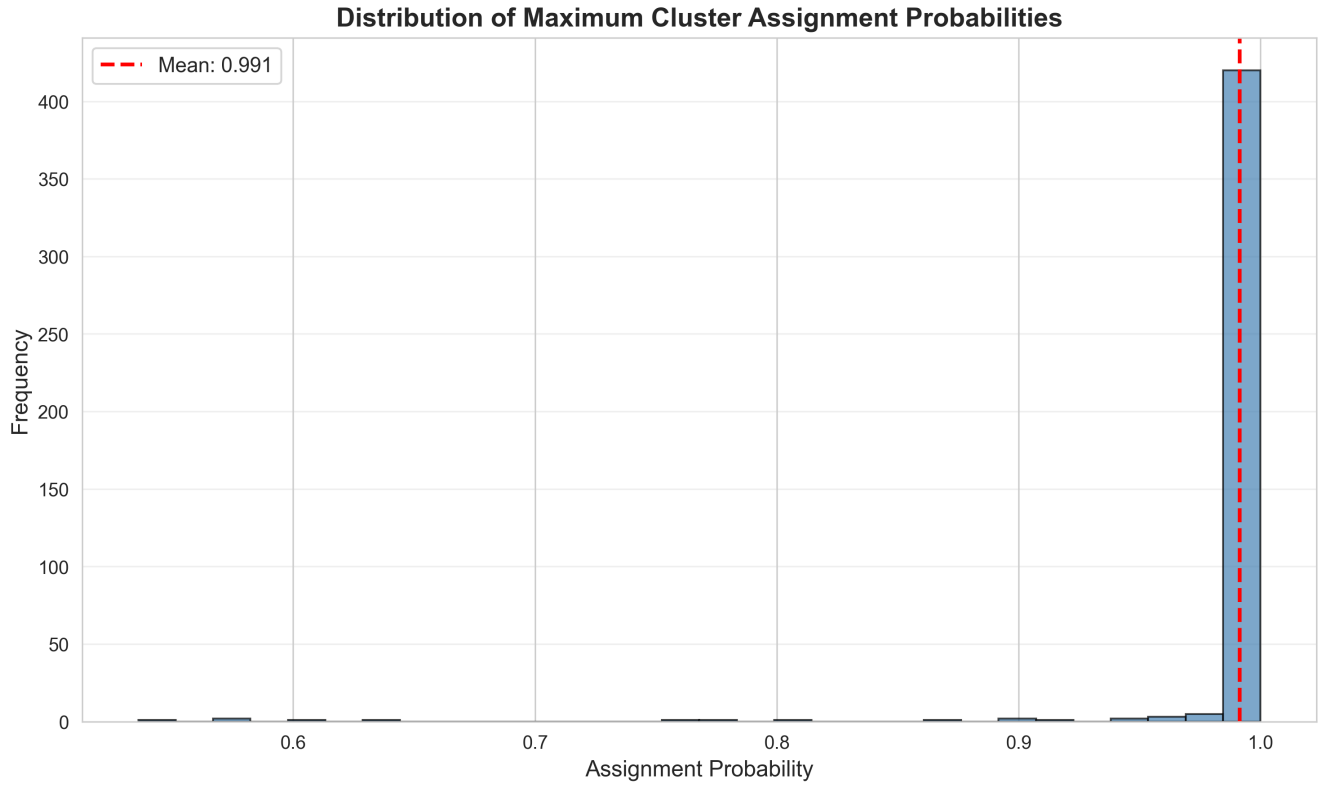

Fig 2: Distribution of maximum cluster assignment probabilities. Mean probability: 0.991, indicating generally confident cluster assignments.

Cluster Distribution:

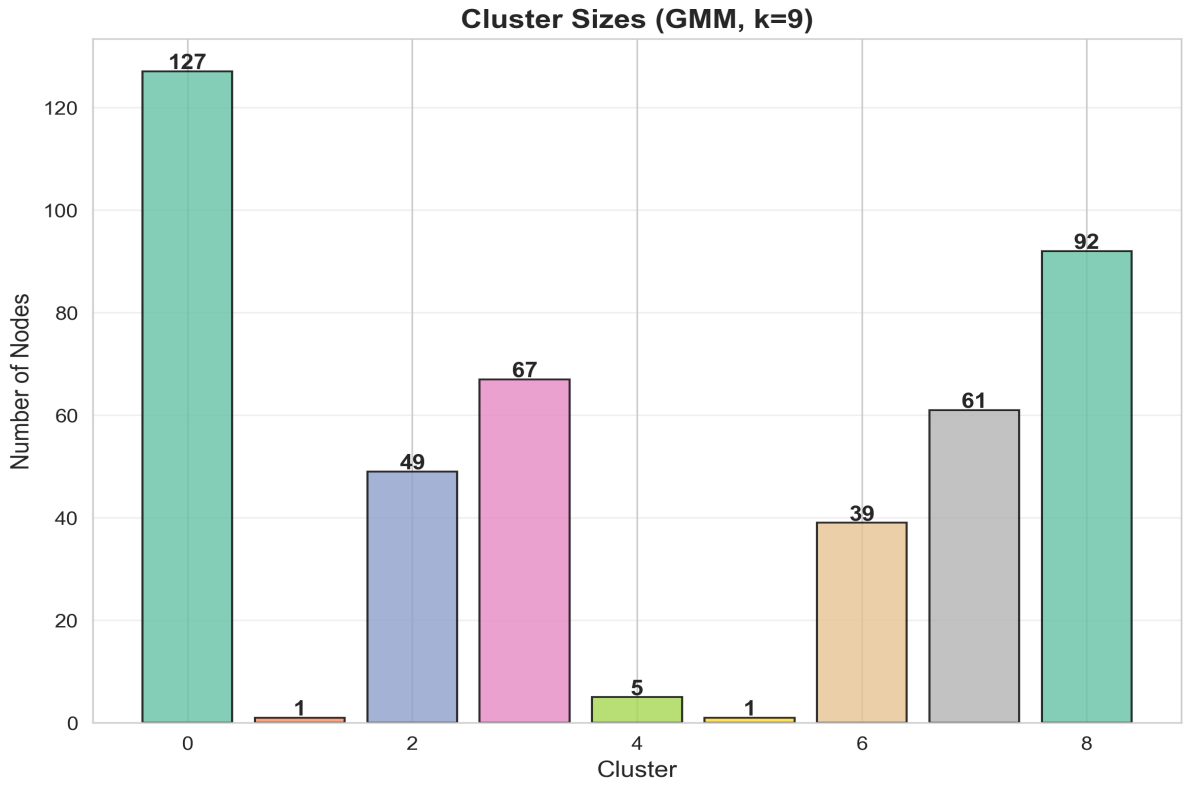

Fig 3: Distribution of nodes across 9 clusters. Cluster sizes reflect the diversity of student roles in the network.

5. Interpretation

The Gaussian Mixture Model identifies distinct student subgroups based on network centrality patterns. Soft probabilities quantify the uncertainty in cluster assignments—students with high probabilities are prototypical cluster members, while those with moderate probabilities occupy boundary regions. The 9-cluster solution balances model complexity (penalized by BIC) with data fit quality.

6. Limitations

GMM assumes multivariate normal distributions within clusters, which may not hold for all datasets. BIC/AIC comparison assumes the models are nested or comparable. Interpretation of cluster probabilities depends on adequate model convergence and initialization.

7. Reproducibility

To reproduce: run the notebook in order from setup through PDF generation. All intermediate figures are saved under `figures/` and clustering results under `gmm_clustering_results.csv`.

Generated on: 2026-01-05 02:51:39